Xiaolong Cheng

Postdoctoral Applicant | LiDAR SLAM, Autonomous UAV Reconstruction, and Multi-Robot Exploration
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RESEARCH PROFILE

Ph.D. candidate in Mechanical Engineering at Southeast University, specializing in LiDAR SLAM, autonomous UAV reconstruction, and multi-robot exploration. My research focuses on robust localization, mapping, and visual–LiDAR fusion for UAV–UGV systems. I develop real-time and cross-platform robotic perception methods that improve geometric registration, environmental adaptability, and system robustness in challenging field scenarios. I am seeking a postdoctoral position in SLAM, autonomous UAV reconstruction, multi-robot exploration, and collaborative robotic systems.

RESEARCH INTERESTS

- LiDAR SLAM and robust localization in challenging environments
- Autonomous UAV-based reconstruction and robotic perception
- Multi-robot exploration, coordination, and cooperative mapping
- Air–ground collaborative SLAM and heterogeneous robotic systems

EDUCATION

Ph.D. Candidate in Mechanical Engineering, Southeast University *09/2021 – 01/2027*

Integrated Master–PhD Program, Nanjing, China
National Scholarship (top 3%)

B.Eng. in Mechanical Engineering, Harbin University of Science and Technology *09/2017 – 06/2021*

Harbin, China
Corporate Scholarship (top 1%) | Outstanding Graduate Student

SELECTED PUBLICATIONS

- **Journal Articles**

[1] K. Geng, **X. Cheng**, P. Ding, K. Shen, T. Ma, and G. Yin, "A Review of Key Technologies in Air-Ground Collaborative Systems" [in Chinese], Journal of Mechanical Engineering ***(CJME),*** doi: 10.3901/JME.260208.
[2] **X. Cheng** et al., "Onion-LO: Why Does LiDAR Odometry Fail Across Different LiDAR Types and Scenarios?," IEEE Robotics and Automation Letters ***(RAL),*** 2025. Code: https://github.com/huashu996/Onion-LO. ***(KITTI Odometry Rank: 26th)***
[3] **X. Cheng** et al., "LIVOX-CAM: Adaptive Coarse-to-Fine Visual-Assisted LiDAR Odometry for Solid-State LiDAR," IEEE Robotics and Automation Letters ***(RAL),*** 2025. Code: https://github.com/huashu996/LIVOX-CAM.
[4] **X. Cheng** et al., "SLBAF-Net: Super-Lightweight Bimodal Adaptive Fusion Network for UAV Detection in Low-Recognition Environment," Multimedia Tools and Applications, 2023. Code: https://github.com/huashu996/SLBAF-Net.

- **Conference Papers**

[5] **X. Cheng**, Y. Sun, K. Geng, T. Ma, and Z. Liu, "Onion-LO++: An Adaptive and Degradation-Resistant Continuous-Time LiDAR Odometry," in Proceedings of the IEEE International Conference on Robotics and Automation ***(ICRA),*** 2026. Code: https://github.com/huashu996/Onion-LO++.
[6] **X. Cheng** et al., "SVM-LO: An Accurate, Robust, Real-Time LiDAR Odometry With Segmentation Voxel Map for Autonomous Vehicles," in 2024 IEEE 27th International Conference on Intelligent Transportation Systems ***(ITSC),*** 2024.
[7] **X. Cheng** et al., "Semantic Mapping Optimization Based on LiDAR and Camera Data Fusion for Autonomous Vehicle," in 2022 6th CAA International Conference on Vehicular Control and Intelligence ***(CVCI),*** 2022. Code: https://github.com/huashu996/camera_lidar_semantic_slam.

- **Under Revision**

[8] **X. Cheng**, K. Geng, C. Ma, Z. Lin, Y. Sun, and T. Ma*, "NDT-CAM: A Simple, Adaptive, and Robust Visual-Assisted LiDAR Odometry for High-Altitude Mapping," Robotics and Autonomous Systems ***(RAS),*** major revision. Code: https://github.com/huashu996/NDT-CAM.

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ACADEMIC SERVICE

- Reviewer for IEEE Robotics and Automation Letters (RA-L)
- Reviewer for ICRA, IROS, and ITSC

RESEARCH EXPERIENCE

Air–Ground Collaborative SLAM | Project Leader *11/2025 – present*

Developed a UAV–UGV collaborative robotic system for cooperative localization and true-color 3D reconstruction, integrating multi-sensor platform calibration, wireless inter-robot communication, and real-time collaborative SLAM for mapping, localization, and autonomous obstacle avoidance.

LiDAR–Vision Fusion-Based Intelligent Road Inspection System | Project Leader *03/2025 – 11/2025*

Developed a public bus multi-sensor mapping system for urban elevated roads, integrating GPS, LiDAR, and vision sensors to support road surface reconstruction, road distress localization, dynamic object removal, and quantitative distress area estimation through LiDAR–camera SLAM and visual detection fusion, resulting in ***publication [3].***

AVP-SLAM Localization System Development | Project Leader *10/2024 – 02/2025*

Led the development of a LiDAR-based autonomous valet parking system using a pre-built global map of a multi-level underground parking garage. The project involved constructing a high-resolution 3D point cloud map, developing a pure LiDAR relocalization algorithm for AVP scenarios, and designing an adaptive LiDAR odometry method to improve robustness across different parking environments, resulting in ***publication [2].***

Mining Area Volume Estimation Based on UAV LiDAR SLAM | Project Leader *02/2024 – 06/2024*

Led the development of a UAV-based LiDAR–camera data acquisition and mapping system for mining-area reconstruction and stockpile volume estimation. The project involved integrating LiDAR and visible-light cameras on a UAV platform, designing a LiDAR–camera–IMU fusion mapping algorithm for accurate 3D reconstruction, and performing point cloud densification, region-of-interest cropping, and volumetric analysis, resulting in ***publication [8].***

Intelligent Vehicle Perception in Low-Visibility Conditions | Core Member *11/2021 – 12/2022*

Led the development of a visible–infrared bimodal perception system for long-range traffic target detection, covering vehicles and pedestrians within 300 m. The project involved building the bimodal sensing platform, collecting and annotating a visible–infrared traffic object detection dataset, and designing the Dual-YOLO bimodal detection network, resulting in ***publication [4].***

TECHNICAL EXPERTISE

Programming: C++, Python, ROS/ROS2, PyTorch, Linux
Libraries/Tools: PCL, OpenCV, Eigen, Ceres Solver, GTSAM, Gazebo, RViz, Docker, Git
Platforms: DJI UAV platforms, UGV, LiDAR, RGB/infrared cameras, GPS/RTK, IMU, NVIDIA Jetson systems

SELECTED HONORS

National Gold Award, Intelligent Driving Challenge, 2022
National Second Prize, 19th Huawei Cup Postgraduate Mathematical Contest in Modeling, 2022
International Meritorious Winner, Mathematical Contest in Modeling (MCM), 2020
National Second Prize, National Undergraduate Mathematical Contest in Modeling, 2019

REFERENCES

Keke Geng, Associate Professor, Southeast University | jsgengke@seu.edu.cn
Guodong Yin, Professor, Southeast University | ygd@seu.edu.cn
Xiaoyuan Zhu, Associate Professor, Southeast University | zhuxy@seu.edu.cn